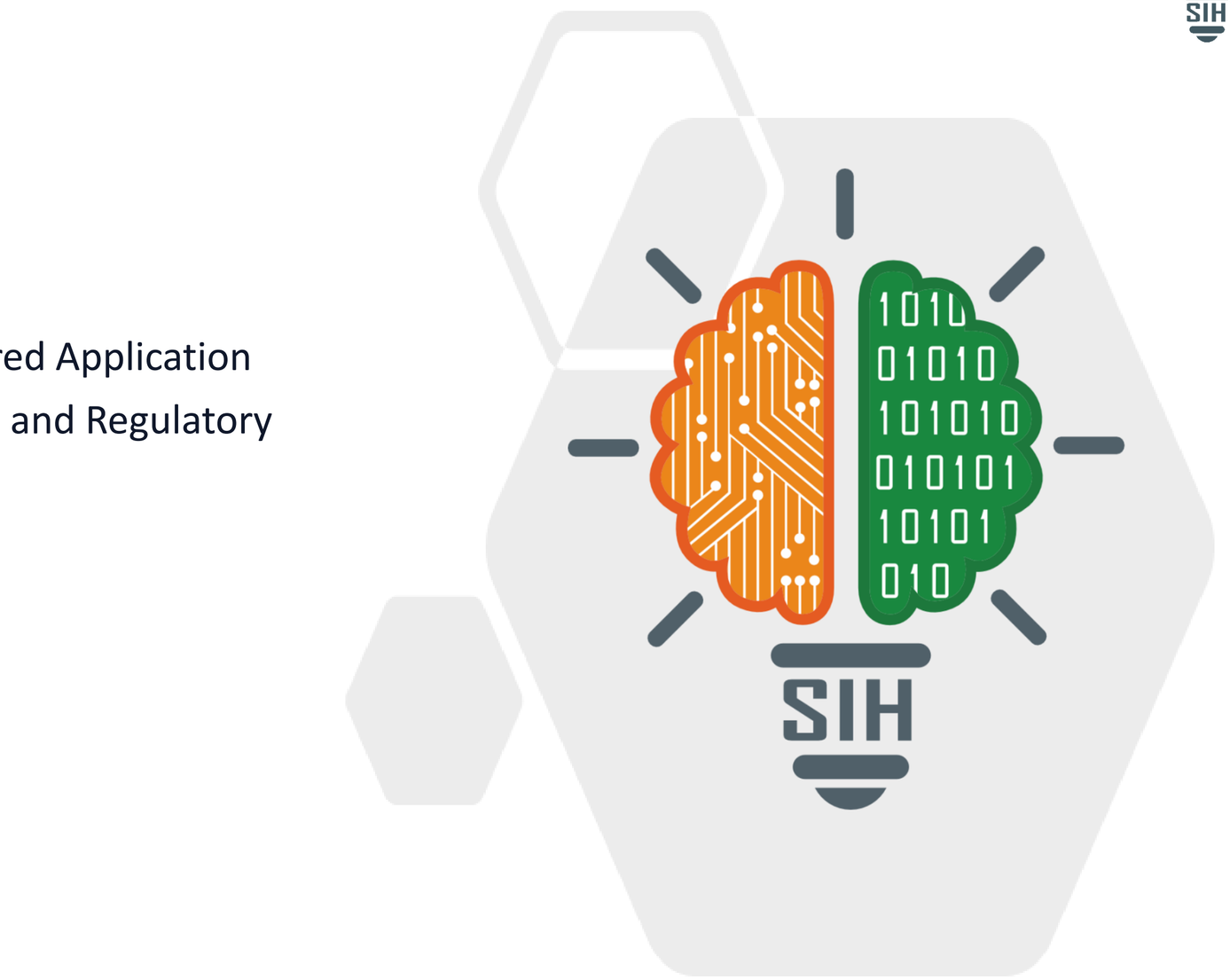
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- **Problem Statement ID:** SIH26261
- **Problem Statement Title:** AI-Powered Application Validation, Document Authentication and Regulatory Compliance Verification
- **Theme:** Smart Automation
- **PS Category:** Software
- **Team ID:**
- **Team Name:** NAVDRISHTI



NAVDRISHTI

ARGUS: Authentication & Recognition Guardian for Unified Screening

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An air-gap-ready compliance and forensic screening engine designed to accelerate institutional approval scrutiny, targeting under 15 minutes per dossier with deterministic rule-based verification.

1. The 4-Pillar Forensic Engine:

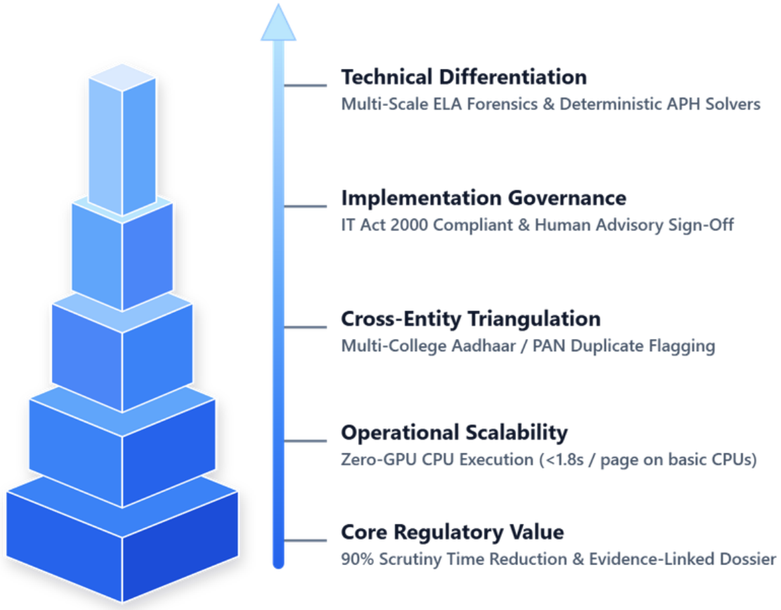
- **Pixel-Level Forensic Scrutiny:** Multi-scale ELA (50/70/90) and noise analysis detect doctored dates, whiteout masking, and cloned municipal seals on Fire NOCs & Land Deeds before OCR extraction.
- **Deterministic APH Norms Solver:** Applies explicit mathematical constraints to FSR, cadre ratios, and carpet-area requirements, producing reproducible calculations and traceable rule-level results.
- **Inter-Document Triangulation:** Graph-based cross-document entity matching detecting duplicate professors across colleges via Aadhaar/PAN hashing and safety capacity mismatches.
- **Cryptographic Trust Bridge:** Decodes official QR seals and validates PAdES/PKCS#7 Digital Signatures offline (with planned DigiLocker & State e-District API sync in connected environments).

2. Core Regulatory Impact:

- **Cuts Scrutiny Time by 90%:** Projected to accelerate the standard 14-day committee scrutiny cycle to under 15 minutes per college.
- **Prevents Approval Slippage:** Catches subtle digital manipulation, expired safety certificates, and unrecorded intake expansions.
- **Evidence-Linked Audit Trail:** Produces a structured deficiency matrix with rule-level statutory citations for objective review by Standing Hearing Committees (SHC).

3. Key Technical Innovations:

- **Decoupled Architecture:** Isolates AI extraction from deterministic math rules for traceable decisions with rule-level audit logs.
- **Forensics First:** Analyzes pixel compression artifacts & EXIF before OCR extraction.
- **Human-in-the-Loop:** Strictly an advisory assistant; final regulatory sign-off remains with AICTE.



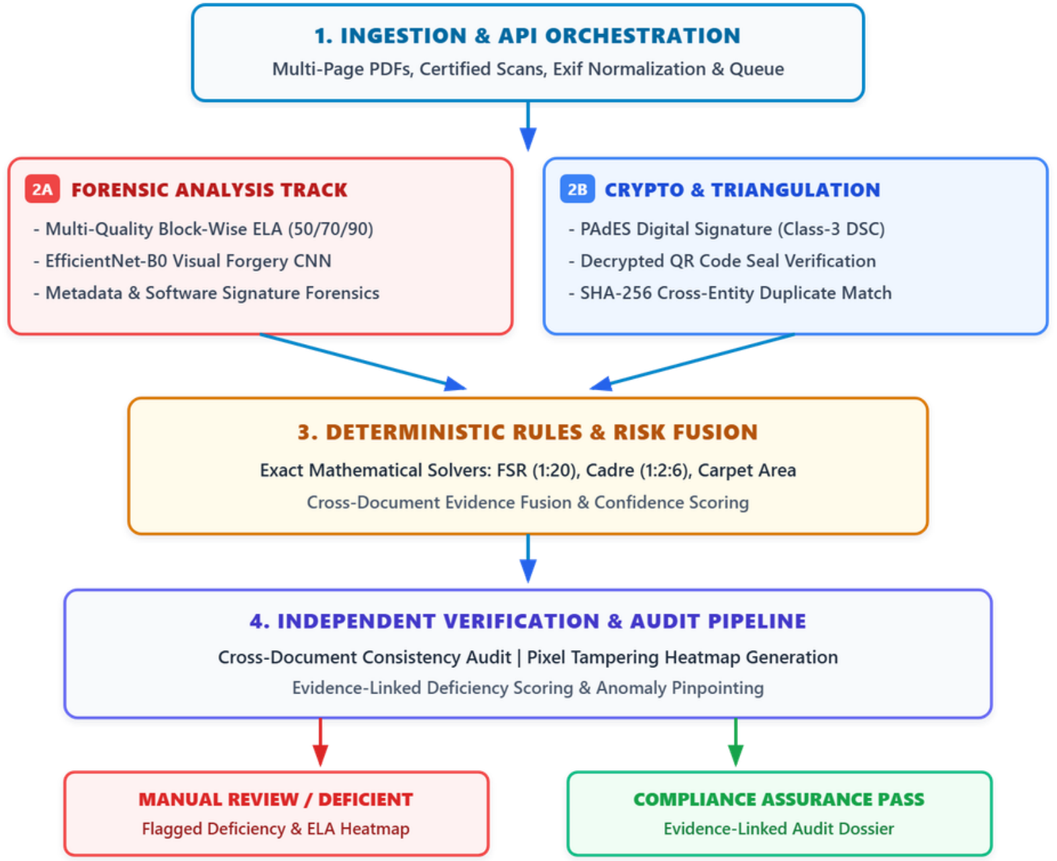
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TECHNICAL APPROACH

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Architectural Layers & Technologies to be Used:

1. **Core Forensics & Vision Layer:** PyTorch (EfficientNet-B0 visual forgery CNN), OpenCV (Multi-scale ELA & morphological filtering), PyHanko (PAdES Digital Signature verification), PyZbar (QR seals).
2. **Ingestion & Extraction:** PyMuPDF (300-DPI page rendering and multi-page processing), EasyOCR (Full-page tables & layout), TrOCR (Reserved strictly for cropped anomalous signatures & dates).
3. **Serving & Scrutiny Cockpit:** FastAPI (Async microservice architecture), Redis (Queue worker), PostgreSQL (Tamper-evident audit logs), React.js (Scrutiny Officer Cockpit with pixel heatmap overlays).
4. **Air-Gapped & On-Premises Ready:** Engineered for secure self-hosted deployment on AICTE servers; zero sensitive institutional data (faculty Aadhaar, financial deeds) is transmitted to third-party commercial APIs.



PyTorch FastAPI OpenCV JS React

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FEASIBILITY AND VIABILITY

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1. Analysis of Feasibility:

- **Technical Feasibility:** Built on standard, battle-tested computer-vision algorithms (ELA, CNN, constraint solvers) requiring no specialized supercomputing hardware.
- **Operational Feasibility:** Seamlessly integrates with existing web portals as an asynchronous REST API microservice without altering institutional submission workflows.
- **Legal & Standards Alignment:** Implements cryptographic verification workflows informed by applicable electronic-signature requirements and relevant PDF signature standards (PAdES / ISO 32000-1).
- **Hardware Footprint:** Operates on standard 4-core cloud/on-prem CPU • < 1.5 GB RAM footprint • Zero GPU dependency.

2. Potential Challenges & Risks:

- **Risk 1 (Document Diversity & Scan Quality):** Low-DPI scans, mobile camera distortions, or wrinkled physical paper uploads causing false alarms.
- **Risk 2 (Adverse Regulatory Misjudgement):** Risk of an AI algorithm incorrectly rejecting an institution's application.
- **Risk 3 (High Submission Volume):** Peak-load bottleneck when thousands of institutions submit EoA applications in April–May.

3. Engineering Mitigation Strategies:

- **Mitigation 1 (Robust Preprocessing):** Uses Adaptive Median Absolute Deviation (MAD) normalization and multi-probe ELA (qualities 50, 70, 90) to help reduce false positives on authentic scanned documents.
- **Mitigation 2 (Strict Advisory Mode):** Outputs flags (VERIFIED, ANOMALY, MANUAL-REVIEW-REQUIRED). Final regulatory action strictly requires human scrutiny officer sign-off.
- **Mitigation 3 (Asynchronous Queue Architecture):** Redis job queuing, stateless horizontal FastAPI workers, and parallelized multi-page chunking (<1.8s/page on 4-core CPU).

Viability & Scale Dimension	Engineering Mitigation Strategy
Infrastructure & Cost Viability	Zero GPU Dependency: Operates on standard 4-core CPU (<1.5 GB RAM); projected ~85% compute cost reduction compared to dedicated cloud GPU clusters (commodity CPU baseline).
Data Sovereignty & Cert-In Compliance	Air-Gapped On-Premises Deployment: Self-hosted inside NIC/AICTE data centers; sensitive PII (PAN/biometrics) never leaves government servers.
Throughput & Peak-Season Elasticity	Projected Target Throughput: ~800 dossiers/day per 4-core node (assumes avg. 200-page dossier; stateless Redis + FastAPI async worker scaling).

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IMPACT AND BENEFITS

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90% Time Reduction

Projected workflow acceleration: Targets reducing institutional dossier scrutiny from a 14-day manual cycle to under 15 minutes per college.

Cross-Institutional Faculty Duplicate Detection

Automated Aadhaar / PAN Duplicate Flagging Rate across institutions via cross-entity graph matching.

Evidence-Linked Audit Trail

Deficiency citations tied directly to statutory APH clauses, supporting objective, defensible review during institutional hearings.

Illustrative Synthetic Dossier — APH Compliance Evaluation

Inspection Parameter	Mandatory APH Norm	Extracted / Computed	Status Flag	Evidence & Statutory Rule
Faculty-to-Student Ratio (FSR)	Max 1 : 20 (B.Tech Engg)	1 : 18.2 (52 Faculty, 946 Students)	● COMPLIANT	APH 2024–27 Clause 7.1
Cadre Ratio (Prof : Assoc : Asst)	1 : 2 : 6 mandatory ratio	0 : 2 : 7 (Zero Professors Found)	● DEFICIENT	APH Norms Table 2.1 (Deficiency)
Fire Safety NOC Expiry	Valid on date of scrutiny	Expired: 12-Dec-2025	● ANOMALY	Page 4 (Municipal Fire NOC Seal)
Approved Intake vs Fire NOC Cap	Fire NOC Cap ≥ Total Intake	NOC Cap: 450 | Total Intake: 600	● SAFETY RISK	Inter-Doc Triangulation (Building Plan)
Faculty Identity Integrity	Unique Aadhaar / PAN / Biometrics	Duplicate in Inst. #1042	● DUPLICATE IDENTITY	Salted HMAC-SHA256 & Face Embedding Match (Advisory Flag)

▲ *Figure: Scrutiny Officer Deficiency Matrix — Actionable cockpit output produced for Standing Hearing Committees (SHC)*

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RESEARCH AND REFERENCES

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1. Regulatory Frameworks & Standards (Statutory Baselines):

• **AICTE Approval Process Handbook (APH 2024–27):** Statutory norms for FSR (1:20), Cadre Ratios (1:2:6), Instructional/Administrative area, and Land/Building norms. [aicteindia.org/bureaus/approval/aph-2024-27]

• Information Technology Act, 2000 (Section 3 & 3A): Statutory criteria for legal validity of electronic records, asymmetric cryptosystems, and Digital Signature Certificates (DSC). [indiacode.nic.in/handle/123456789/1999]

• **ISO 32000-1 / PAdES Standard (ETSI TS 102 778):** International standards for PDF Advanced Electronic Signatures, timestamping, and long-term document integrity validation. [iso.org/standard/51502.html]

2. Academic & Forensic Research References:

• **Error Level Analysis (ELA) in Digital Forensics:** Krawetz, N. (2007), "A Picture's Worth... Digital Image Analysis & JPEG Compression Artifacts", Black Hat Briefings. [hackerfactor.com/papers/krawetz-ela.pdf]

• Convolutional Neural Networks for Image Forgery: Tan, M., & Le, Q. V. (2019), "EfficientNet: Rethinking Model Scaling for CNNs", ICML (arXiv:1905.11946). [arxiv.org/abs/1905.11946]

• **Benchmark Datasets in Document Forensics:** CASIA Image Tampering Dataset v2.0 (CASIA-TIDE) & RVL-CDIP Document Classification Dataset. [forensics.idealtest.org/casiav2]

• **Digital Public Infrastructure (DPI) & Verification APIs:** National Academic Depository (NAD) & NIC DigiLocker Developer API Integration Specifications. [partners.digitallocker.gov.in]

Live Proof of Concept (Working Prototype)

- **GitHub Repository:** ARGUS-GitHub[https://github.com/Aravkataria/ARGUS]
- **Validated Core Features:** Multi-scale ELA + EfficientNet-B0 visual forensics; 300-DPI multi-page OCR table parsing; deterministic Python math solvers for FSR, Cadre & Fire NOC; evidence-linked deficiency reports.
- • **Prototype Execution Specs:** Lightweight CPU runtime (<1.8s/page on 4-core CPU); 100% air-gapped on-premises architecture; designed to compress 14-day committee review to < 15 mins.

